

The Non-Linear Impact Of Leverage On Firm Performance: A Panel Data Analysis Of Indian Manufacturing Firms



Subject: Panel Data Analysis

**Topic: The Non-Linear Impact Of Leverage On Firm
Performance: A Panel Data Analysis Of Indian Manufacturing
Firms**

Submitted By,

PRAKALYA GANESH

PES1PG25MB383

TRIMISTER 2

DEPARTMENT OF MBA

Submitted To,

Prof . AJAY MASSAND

DEPARTMENT OF MBA

PES UNIVERSITY.

ABSTRACT

This study examines the relationship between financial leverage and firm performance among BSE-listed Indian manufacturing firms, with a particular focus on whether this relationship is non-linear. It uses a balanced panel dataset covering a ten-year period that includes both relatively stable economic conditions and major disruptions such as the COVID-19 pandemic and the subsequent monetary tightening cycle. Drawing on Trade-off Theory and Pecking Order Theory, the analysis is carried out using a Two-Way Fixed Effects panel regression model with firm-level clustered standard errors, supported by the Hausman test for model selection. The results show that leverage has a consistently negative effect on firm profitability, and there is no clear evidence to support the inverted-U relationship predicted by Trade-off Theory. Overall, the study adds to the literature on capital structure in emerging markets by providing one of the first direct tests of non-linearity in the leverage-performance relationship for Indian manufacturing firms using recent post-COVID data.

Keywords: Capital Structure, Financial Leverage, Firm Performance, Panel Data, Indian Manufacturing

JEL Classification: G32, L60, C23

Acknowledgements: *Data sourced from the Capitaline corporate database*

1. INTRODUCTION

The question of how a firm's capital structure affects its financial performance remains one of the most widely debated issues in corporate finance. The discussion dates back to the work of Modigliani and Miller (1958), who argued that in perfectly efficient capital markets, a firm's value is independent of how it chooses to finance itself. Over time, however, researchers began to challenge this idea by introducing real-world factors that make capital structure decisions matter.

One of the earliest extensions came from Modigliani and Miller (1963), who incorporated corporate taxes into the model and showed that debt can create value through the tax shield it provides. This insight laid the foundation for Trade-off Theory, which suggests that firms try

The Non-Linear Impact Of Leverage On Firm Performance: A Panel Data Analysis Of Indian Manufacturing Firms

to strike a balance between the benefits of debt, mainly tax savings, and the rising costs associated with financial distress. According to this view, firms should have an optimal level of leverage, beyond which additional borrowing begins to reduce rather than enhance firm value.

At the same time, an alternative perspective emerged. Myers (1984) and Myers and Majluf (1984) proposed the Pecking Order Theory, which argues that firms follow a financing hierarchy rather than targeting an optimal capital structure. Because of information asymmetry between managers and external investors, firms prefer to rely on internal funds first, then debt, and issue equity only as a last option. This approach implies that higher leverage may not necessarily reflect value-maximising behavior, but instead financial constraints.

Adding another layer to this discussion, Jensen and Meckling (1976) introduced agency cost theory, which focuses on conflicts between managers and shareholders. They argued that debt can help discipline managers by limiting the amount of free cash flow available for discretionary spending. Jensen (1986) later extended this idea, suggesting that higher leverage can reduce managerial inefficiency by forcing firms to commit to fixed financial obligations. Taken together, these theories — brought together comprehensively by Harris and Raviv (1991) — offer different and sometimes conflicting predictions about how leverage should affect firm performance. This theoretical ambiguity has led to a large body of empirical work attempting to test these relationships.

Empirical findings, however, remain far from consistent. In cross-country research, Rajan and Zingales (1995) found that leverage is generally negatively associated with profitability across G7 countries, while also identifying firm size, tangibility, and growth opportunities as key determinants that must be accounted for in empirical models. Studies focusing on emerging markets show similar patterns. For instance, Zeitun and Tian (2007) reported a significantly negative relationship between capital structure and firm performance in Jordan, while Ibhagui and Olokoyo (2018) found that this negative effect varies depending on firm size in the Nigerian context. On the methodological side, Margaritis and Psillaki (2010) made an important contribution by testing for non-linearity in the leverage-performance relationship, showing that the effect of debt is not necessarily linear. Their findings highlight the limitation of models that assume a simple linear relationship between leverage and performance.

When it comes to India, the evidence is still relatively limited and somewhat mixed. Chadha and Sharma (2015), studying BSE-listed manufacturing firms, found that leverage does not significantly affect ROA, whereas Singh and Bagga (2019) reported a positive relationship between capital structure and profitability for firms in the Nifty 50 index. More recent studies by Panchal and Chand (2023) and Pal (2022) point in the opposite direction, documenting a significant negative relationship between leverage and firm performance. Pal (2022) further notes that this negative impact becomes more pronounced during periods of tighter credit conditions. Despite these contributions, an important gap remains. Most studies on Indian manufacturing firms have assumed a linear relationship between leverage and performance, without testing the inverted-U pattern predicted by Trade-off Theory. As a result, the question of whether an optimal level of leverage actually exists in this context has not been adequately addressed.

This study aims to fill that gap. Using a balanced panel of 78 BSE-listed Indian manufacturing firms over the period 2016 to 2025 — a timeframe that captures different economic phases including pre-COVID stability, the pandemic shock, and the subsequent monetary tightening by the Reserve Bank of India in 2022–2023 — the study examines whether the leverage-performance relationship is non-linear. By incorporating both the leverage variable and its squared term in a Two-Way Fixed Effects regression framework with firm-level clustered standard errors, the analysis directly tests the key prediction of Trade-off Theory. At the same time, it also evaluates whether the observed relationship aligns more closely with the Pecking Order Theory. In addition, sub-period analysis is used to assess whether the relationship changed after COVID-19. The results are expected to offer useful insights for corporate decision-makers as well as policymakers concerned with the financial health and debt sustainability of India's manufacturing sector.

2. LITERATURE REVIEW

1. Modigliani, F., & Miller, M. H. (1958). "The Cost of Capital, Corporation Finance and the Theory of Investment." *The American Economic Review*, 48(3), 261–297. This paper forms the starting point of modern capital structure theory. It puts forward the irrelevance theorem, arguing that in a perfect market with no taxes or frictions, a firm's value

does not depend on how it is financed. Later theories, including the ones used in this study, essentially build on this framework by relaxing its assumptions.

2. Modigliani, F., & Miller, M. H. (1963). "Corporate Income Taxes and the Cost of Capital: A Correction." The American Economic Review, 53(3), 433–443.

In this follow-up paper, the authors introduced corporate taxes into their earlier model and showed that debt creates value through tax savings. This idea of a “tax shield” becomes central to Trade-off Theory and directly supports the argument that leverage can positively influence firm performance under certain conditions.

3. Myers, S. C. (1984). "The Capital Structure Puzzle." The Journal of Finance, 39(3), 575–592.

Myers discussed why firms do not behave exactly as traditional theories predict and introduced what later became known as the Pecking Order Theory. He argued that firms prefer using internal funds first, then debt, and only issue equity as a last resort due to information differences between managers and investors. This perspective helps explain why high leverage can often be linked to weaker performance.

4. Myers, S. C., & Majluf, N. S. (1984). "Corporate Financing and Investment Decisions When Firms Have Information That Investors Do Not Have." Journal of Financial Economics, 13(2), 187–221.

This paper builds on the idea of information asymmetry by showing how it affects financing decisions. It explains why external financing becomes costly and why firms tend to avoid issuing equity. This logic forms the backbone of the Pecking Order Theory and supports the interpretation of results where debt is not necessarily beneficial for firm performance.

5. Jensen, M. C., & Meckling, W. H. (1976). "Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure." Journal of Financial Economics, 3(4), 305–360.

Here, the concept of agency costs is introduced. The authors argue that managers may not always act in the best interest of shareholders, and that debt can act as a disciplining mechanism by limiting excess spending. This provides a theoretical reason why moderate levels of leverage could improve firm performance.

6. Jensen, M. C. (1986). "Agency Costs of Free Cash Flow, Corporate Finance, and Takeovers." The American Economic Review, 76(2), 323–329.

Jensen extends his earlier work by focusing on free cash flow. He suggests that higher debt forces firms to use their cash more efficiently by committing to fixed interest payments. This adds nuance to the discussion by showing that debt can both help and hurt performance depending on the situation.

7. Harris, M., & Raviv, A. (1991). "The Theory of Capital Structure." *The Journal of Finance*, 46(1), 297–355.

This paper provides a comprehensive review of the main theories of capital structure, including tax-based, agency, and information-based explanations. It is often used to outline the broader theoretical landscape and to understand the different predictions each theory makes regarding leverage and firm value.

8. Margaritis, D., & Psillaki, M. (2010). "Capital Structure, Equity Ownership and Firm Performance." *Journal of Banking & Finance*, 34(3), 621–632.

Based on data from French manufacturing firms, this study finds that leverage can improve performance up to a certain point. It is especially relevant because it uses a quadratic specification to test for non-linearity, a method that is also applied in this study to examine whether an optimal debt level exists.

9. Chadha, S., & Sharma, A. K. (2015). "Capital Structure and Firm Performance: Empirical Evidence from India." *Vision: The Journal of Business Perspective*, 19(4), 295–302.

This study looks at Indian manufacturing firms and finds mixed results — leverage does not significantly affect ROA or Tobin's Q, but it negatively impacts ROE. It serves as an important benchmark within the Indian context, especially since it uses a similar sector and dataset approach.

10. Panchal, S., & Chand, S. (2023). "Impact of Capital Structure on Firm Performance: Empirical Evidence from Indian Manufacturing Sector." *Journal of Asia Business Studies*, 17(5), 1063–1080.

Using a more recent dataset, this paper finds that higher debt levels are linked to lower profitability, particularly in terms of ROA. Its findings are closely aligned with those of this study and provide strong contemporary support for a negative leverage-performance relationship.

11. Singh, N. P., & Bagga, M. (2019). "The Effect of Capital Structure on Profitability: An Empirical Panel Data Study." *Jindal Journal of Business Research*, 8(1), 65–77.

This study reports a positive relationship between leverage and profitability, which contrasts with many other findings. Including it highlights that results can vary depending on the sample, time period, and methodology used, showing that the debate in this area is still open.

12. Zeitun, R., & Tian, G. G. (2007). "Capital Structure and Corporate Performance: Evidence from Jordan." *Australasian Accounting, Business and Finance Journal*, 1(4), 40–61.

This paper, based on firms in Jordan, finds that higher leverage leads to lower firm performance. It is widely cited in emerging market research and helps place the findings of this study in a broader developing economy context.

13. Ibhagui, O. W., & Olokoyo, F. O. (2018). "Leverage and Firm Performance: New Evidence on the Role of Firm Size." *North American Journal of Economics and Finance*, 45, 57–82.

This study shows that the impact of leverage depends on firm size, with smaller firms suffering more from higher debt levels. It adds an important dimension by suggesting that leverage effects are not uniform across firms, which is relevant for interpreting robustness checks involving firm size.

14. Pal, S. (2022). "Influence of Capital Structure on Firm Performance: Empirical Evidence from Indian Manufacturing Industry." *International Journal of Business Management and Finance Research*, 5(2), 100–110.

This research finds a negative relationship between leverage and profitability in Indian manufacturing firms, especially during tight credit periods. It closely supports the post-COVID findings of this study and strengthens the argument of a context-dependent leverage effect.

15. Rajan, R. G., & Zingales, L. (1995). "What Do We Know About Capital Structure? Some Evidence from International Data." *The Journal of Finance*, 50(5), 1421–1460.

This well-known cross-country study identifies firm size, tangibility, profitability, and growth as key determinants of capital structure. It provides the empirical foundation for including similar control variables in this study.

3. RESEARCH OBJECTIVES

- To examine the nature and direction of the relationship between financial leverage and firm performance among BSE-listed Indian manufacturing firms during the period 2016 to 2025.
- To test empirically whether this relationship is non-linear — in particular, whether there exists an optimal level of debt beyond which firm profitability begins to fall, as suggested by Trade-off Theory.
- To identify firm-level characteristics that significantly influence profitability after accounting for capital structure decisions, using a Two-Way Fixed Effects panel framework.
- To explore whether the relationship between leverage and performance changed structurally in the post-COVID-19 period, given the disruptions in credit markets and the monetary tightening cycle initiated by the Reserve Bank of India.
- To derive policy-relevant insights on optimal capital structure decisions for Indian manufacturing firms operating under different macroeconomic and interest rate conditions.

4. HYPOTHESES

The hypotheses are drawn from two major and competing frameworks in capital structure literature — the **Trade-off Theory** and the **Pecking Order Theory** — and align with the research objectives above.

H1 — The Leverage–Performance Relationship (Directionality)

H_{1o} : Financial leverage has no statistically significant effect on the Return on Assets of Indian manufacturing firms.

H_{1a} : Financial leverage has a statistically significant effect on the Return on Assets of Indian manufacturing firms.

The Non-Linear Impact Of Leverage On Firm Performance: A Panel Data Analysis Of Indian Manufacturing Firms

Theoretical basis: Both Trade-off Theory and Pecking Order Theory predict that leverage and performance are linked, although they differ in how this relationship behaves. This hypothesis focuses on establishing whether a relationship exists before examining its form.

H2 — Non-linearity and the Existence of an Optimal Leverage Point (Core Hypothesis)

H₂₀: The relationship between financial leverage and firm performance is purely linear, meaning no significant quadratic effect is present.

H_{2a}: The relationship between financial leverage and firm performance is non-linear (inverted-U shaped), implying the presence of an optimal level of leverage beyond which profitability declines, consistent with Trade-off Theory.

Theoretical basis: Trade-off Theory (Modigliani and Miller, 1963; Myers, 1984) suggests that firms balance the tax advantages of debt against increasing financial distress costs, leading to an optimal leverage level. A significant coefficient on the squared leverage term (LEV^2) in the regression would provide evidence for this.

H3 — Firm-Level Control Variables and Performance

H₃₀: Firm size, asset tangibility, sales growth and liquidity do not significantly affect profitability once leverage is accounted for.

H_{3a}: At least one of the control variables — firm size, asset tangibility, sales growth or liquidity — significantly influences profitability after accounting for leverage.

Theoretical basis: Prior research in corporate finance and resource-based theory consistently finds these variables to be important drivers of firm performance. Including them helps reduce omitted variable bias and improves the reliability of leverage estimates.

H4 — Structural Break: Pre- vs Post-COVID Leverage Effect

H₄₀: There is no meaningful difference in the leverage-performance relationship between the pre-COVID (2016–2019) and post-COVID (2020–2025) periods.

H_{4a}: The leverage-performance relationship changes significantly in the post-COVID period due to financial stress and the rise in borrowing costs following monetary tightening.

Theoretical basis: The RBI increased repo rates by 250 basis points between May 2022 and February 2023, raising borrowing costs for firms. Under such conditions, the advantages of leverage predicted by Trade-off Theory are likely to weaken, leading to a more consistently negative effect.

5. RESEARCH METHODOLOGY

5.1 Research Design

This study follows a **quantitative and explanatory research design**, relying on secondary panel data. The aim is to understand the direction, magnitude and shape of the relationship between capital structure and firm performance. These questions are best addressed using panel econometric techniques rather than cross-sectional or case-based methods.

Using panel data over a ten-year period allows the analysis to focus on changes within firms over time, while controlling for stable firm-specific characteristics through fixed effects.

5.2 Data Source and Sample Selection

The data is sourced from the **Capitaline corporate database**, which provides detailed financial information on Indian firms. The analysis focuses on **BSE-listed manufacturing firms**, as this sector is asset-heavy, relies more on debt financing and is sensitive to interest rate movements, making it suitable for testing capital structure theories.

The final sample includes **78 Manufacturing firms** observed annually from **2016 to 2025**, resulting in a balanced panel of **780 firm-year observations**. The following criteria were used to construct the sample:

Firms had to report complete data for all variables across the entire study period to maintain a balanced panel. Financial firms and utilities were excluded since their capital structures are largely influenced by regulation. Micro-cap firms with total assets below ₹50 crores were also excluded due to concerns over data quality. Firms with unclear sector classifications were removed to ensure consistency within the sample.

5.3 Variables

Dependent

Variable:

Return on Assets (ROA), defined as Net Profit divided by Total Assets, is used as the measure of firm performance. It is preferred over market-based indicators like Tobin’s Q because it reflects operational efficiency without being influenced by market sentiment.

Independent

Variables:

The main independent variable is **LEV** (Total Debt divided by Total Assets), which captures leverage. Its squared term **LEV²** is included to test for non-linear effects. This approach is commonly used in empirical studies to examine whether an optimal debt level exists (Margaritis and Psillaki, 2010).

Control

Variables:

Four control variables are included. **Firm Size** (log of Total Assets) accounts for scale-related effects. **Tangibility** (fixed assets plus capital work-in-progress divided by total assets) captures asset structure and collateral value. **Sales Growth** (year-on-year change in sales) reflects demand-side momentum. **Liquidity** (current ratio) captures short-term financial health and its interaction with debt capacity.

5.4 Data Treatment

Before estimation, all ratio variables were **winsorized at the 1st and 99th percentiles** to limit the impact of extreme values. Some observations showed leverage above 1.0, indicating firms with liabilities exceeding assets; these were adjusted within the winsorized range.

Two observations with missing sales growth data (Sanco Industries, 2023 and 2025) were dropped automatically during estimation, leaving **778 usable observations**.

5.5 Estimation Strategy

The analysis follows a four-stage approach:

Stage

1

—

Pooled

OLS:

A baseline regression is estimated under the assumption that no firm-specific unobserved effects exist. This serves only as a reference point and is expected to be biased.

Stage 2 — Hausman Test:

This test determines whether fixed or random effects should be used by checking whether firm-specific effects are correlated with the explanatory variables. If the null is rejected, fixed effects are preferred.

Stage 3 — Two-Way Fixed Effects (TWFE):

This forms the main model. It includes firm fixed effects to control for time-invariant firm characteristics and year fixed effects to capture common macroeconomic shocks. Standard errors are clustered at the firm level to account for heteroskedasticity and serial correlation.

Stage	4	—	Robustness	Checks:
-------	---	---	------------	---------

The TWFE model is re-estimated for two sub-periods (2016–2019 and 2020–2025) and for a sample excluding extreme firm sizes. Stability in results across these variations indicates robustness.

5.6 Diagnostic Testing

Three diagnostic tests are used after estimation. The **Breusch-Pagan test** checks for heteroskedasticity. The **Durbin-Watson statistic** is used to detect serial correlation. The **Pesaran test** assesses cross-sectional dependence across firms.

Where any issues are detected, clustered standard errors are used to correct them rather than changing the model coefficients themselves.

5.7 Software

All analysis was carried out in **Jupyter Notebook**. The pandas and numpy libraries were used for data handling, statsmodels for OLS estimation and diagnostics, and linearmodels for fixed effects panel regression with clustered errors. The regression outputs were then formatted into publication-style tables for presentation.

6. DATA ANALYSIS, RESULTS AND INTERPRETATION

6.1 Sample and Panel Structure

The Non-Linear Impact Of Leverage On Firm Performance: A Panel Data Analysis Of Indian Manufacturing Firms

The dataset consists of a balanced panel of **78 BSE-listed Indian manufacturing firms** covering the period from **2016 to 2025**, resulting in a total of **780 firm-year observations**. The data was obtained from the Capitaline corporate database. Since each firm contributes exactly 10 annual observations, the panel remains fully balanced, which helps eliminate concerns about attrition bias. All financial variables are expressed in Indian Rupees (₹ Crores) and converted into ratios before estimation.

To reduce the influence of extreme values, all ratio variables were **winsorized at the 1st and 99th percentiles**. For example, the raw leverage (LEV) variable had a maximum value of 1.118, indicating that some firms held debt exceeding their total assets, a clear sign of financial distress. After winsorization, LEV was capped at 0.685. Similarly, the minimum value of ROA was limited to -0.016, removing the effect of extreme losses.

Two firm-year observations for Sanco Industries (2023 and 2025) had missing sales growth values and were automatically dropped from the regression by the panel estimator. As a result, all regression models are based on **778 usable observations**.

6.2 Variable Construction

All regression variables were derived from the raw Capitaline dataset. The dependent variable, **Return on Assets (ROA)**, is calculated as Net Profit divided by Total Assets and reflects operating profitability. The key independent variable, **LEV**, is defined as Total Debt divided by Total Assets and captures leverage.

To test for non-linear effects predicted by Trade-off Theory, the squared term **LEV²** is included. Control variables include **Firm Size** (log of Total Assets), **Tangibility** (Net Fixed Assets plus Capital Work-in-Progress divided by Total Assets), **Sales Growth** (year-on-year percentage change in Net Sales) and **Liquidity** (Current Ratio). All estimations use the winsorized versions of LEV, LEV² and ROA.

6.3 Descriptive Statistics

Table 1: Descriptive Statistics

Variable	N	Mean	Median	Std. Dev.	Min	Max
ROA (winsorized)	780	0.1231	0.1194	0.0727	-0.0159	0.3586

The Non-Linear Impact Of Leverage On Firm Performance: A Panel Data Analysis Of Indian Manufacturing Firms

LEV (winsorized)	780	0.1556	0.0847	0.1697	0.0000	0.6848
LEV² (winsorized)	780	0.0530	0.0072	0.0886	0.0000	0.4690
Firm Size	780	3.4627	3.4706	0.5910	1.7017	4.9783
Tangibility	780	0.4247	0.3946	0.2335	0.0179	1.3244
Sales Growth	778	0.0933	0.0923	0.2018	−1.0000	1.3650
Liquidity	780	1.7043	1.4500	0.8768	0.0300	7.5700

A few patterns stand out. The mean ROA of **12.31%** shows that firms in the sample are generally profitable. The closeness between mean and median suggests the distribution is fairly balanced.

The average leverage of **15.56%** is moderate, though the higher standard deviation reflects wide differences across firms. The fact that the median leverage (8.47%) is much lower than the mean indicates a right-skewed distribution, meaning most firms use little debt while a few are heavily leveraged.

Mean tangibility at **42.47%** aligns with the capital-intensive nature of manufacturing. Meanwhile, average sales growth of **9.33%** reflects steady expansion, though the minimum of −100% captures major disruptions during the COVID-19 period.

6.4 Correlation Analysis and Multicollinearity

Table 2: Correlation Matrix

	LEV	LEV²	Firm Size	Tangibility	Sales Growth	Liquidity
LEV	1.000	0.943	−0.163	0.282	−0.075	−0.542
LEV²	0.943	1.000	−0.192	0.215	−0.120	−0.445
Firm Size	−0.163	−0.192	1.000	−0.293	0.027	0.129
Tangibility	0.282	0.215	−0.293	1.000	0.094	−0.401
Sales Growth	−0.075	−0.120	0.027	0.094	1.000	0.009
Liquidity	−0.542	−0.445	0.129	−0.401	0.009	1.000

The high correlation of 0.943 between LEV and LEV² is expected since one is derived from the other. This is a case of intentional multicollinearity and is standard in quadratic models.

The Non-Linear Impact Of Leverage On Firm Performance: A Panel Data Analysis Of Indian Manufacturing Firms

Variance Inflation Factor (VIF) checks were run in two stages. When LEV^2 was excluded, all VIF values were below 2.0, confirming no issue among control variables. When LEV^2 was included, VIF values for LEV and LEV^2 rose above 10, which is expected and not problematic here. All other variables remained well within acceptable limits.

The strong negative correlation between LEV and Liquidity (-0.542) makes economic sense, as firms with higher debt often maintain lower liquidity due to repayment pressures.

6.5 Baseline Pooled OLS

Table 3: Pooled OLS Baseline Regression (Dependent Variable: ROA)

Variable	Coefficient	Std. Error	t-statistic	p-value
Constant	0.1797	0.018	9.944	0.000***
LEV	-0.2584	0.044	-5.846	0.000***
LEV ²	0.1791	0.081	2.204	0.028**
Firm Size	-0.0180	0.004	-4.483	0.000***
Tangibility	0.0245	0.011	2.220	0.027**
Sales Growth	0.0769	0.011	6.884	0.000***
Liquidity	0.0112	0.003	3.459	0.001***
Adj. R ²	0.279			
F-statistic	51.16	p = 0.000		

The pooled OLS model is statistically significant and explains about **27.9%** of the variation in ROA. Both LEV and LEV^2 are significant, suggesting a U-shaped relationship in this simple setup.

However, this model ignores firm-specific characteristics like management quality or reputation, which may influence both leverage and performance. The low Durbin–Watson statistic (0.840) also indicates strong serial correlation, making OLS unreliable. Panel methods are therefore more appropriate.

6.6 Hausman Test and Model Selection

The Non-Linear Impact Of Leverage On Firm Performance: A Panel Data Analysis Of Indian Manufacturing Firms

The Hausman test result ($\chi^2(7) = 53.86$, $p = 0.0000$) rejects the null hypothesis that random effects are appropriate. This means firm-specific factors are correlated with the regressors. As a result, **Fixed Effects estimation** is selected.

6.7 Two-Way Fixed Effects (TWFE) Panel Regression — Main Results

The main model uses firm and year fixed effects, along with clustered standard errors. The estimated model takes the form:

$$\text{ROA}_{it} = \beta_0 + \beta_1 \text{LEV}_{it} + \beta_2 \text{LEV}^2_{it} + \beta_3 \text{SIZE}_{it} + \beta_4 \text{TANG}_{it} + \beta_5 \text{GROWTH}_{it} + \beta_6 \text{LIQ}_{it} + \alpha_i + \lambda_t + \varepsilon_{it}$$

Table 4: Two-Way Fixed Effects Regression Results

Variable	Coefficient	Clustered Std. Error	t-statistic	p-value
Constant	0.5421	0.1145	4.733	0.000***
LEV	-0.1679	0.0505	-3.327	0.001***
LEV ²	-0.0778	0.0760	-1.023	0.307
Firm Size	-0.1038	0.0333	-3.118	0.002***
Tangibility	-0.1116	0.0325	-3.434	0.001***
Sales Growth	0.0810	0.0134	6.058	0.000***
Liquidity	0.0065	0.0066	0.987	0.324
R ² (Within)	0.2481			

6.8 Interpretation of Main Results

Leverage (LEV):

LEV is negative and highly significant. A one-unit increase leads to a **16.79 percentage point drop in ROA**, supporting the idea that debt burdens outweigh benefits.

LEV²:

The squared term is insignificant, meaning there is **no evidence of a non-linear relationship**. The data suggests a consistently negative effect rather than an optimal debt level.

The Non-Linear Impact Of Leverage On Firm Performance: A Panel Data Analysis Of Indian Manufacturing Firms

Firm **Size:**

The negative coefficient reflects **diminishing returns to scale** within firms as they grow larger.

Tangibility:

Higher asset intensity is associated with lower ROA, likely due to costs like depreciation and delayed returns on investment.

Sales **Growth:**

Strong positive effect, confirming that revenue expansion directly improves profitability.

Liquidity:

Not significant, likely because its effect overlaps with leverage.

6.9 Diagnostic Tests

Key diagnostic results show heteroskedasticity and mild serial correlation, both handled through clustered standard errors. Cross-sectional dependence is negligible.

Test	Statistic	p-value	Conclusion
Breusch-Pagan (Heteroskedasticity)	LM = 15.84	0.0146	Heteroskedasticity present
Durbin-Watson (Serial Correlation)	DW = 1.480	—	Positive autocorrelation indicated
Pesaran CD (Cross-sectional Dependence)	Avg. ρ = -0.011	—	Negligible cross-sectional dependence

The Breusch-Pagan test rejects the null hypothesis of homoscedastic errors ($p = 0.015$), confirming the presence of heteroscedasticity in the residuals. The Durbin-Watson statistic of 1.48 — below the benchmark value of 2.0 — is indicative of mild positive serial correlation within firm clusters. Both conditions are standard in firm-level panel data and do not invalidate the estimates, as the main model employs **firm-level clustered standard errors**, which are robust to both heteroscedasticity and within-cluster serial correlation simultaneously. The Pesaran average residual correlation of -0.011 indicates negligible cross-sectional dependence among firm residuals, suggesting that the year fixed effects have successfully absorbed common macro shocks affecting firms simultaneously

6.10 Robustness Analysis

To assess the stability of the core findings, the main TWFE model was re-estimated across two sub-periods and an outlier-restricted sample. Results are presented in Table 6.

Table 6: Robustness Check Results

Variable	Pre-COVID (2016–2019)	Post-COVID (2020–2025)	Excl. Outliers	Size
LEV	0.0279 (0.762)	−0.2084** (0.033)	−0.1712*** (0.002)	
LEV ²	−0.2692** (0.047)	−0.0055 (0.971)	−0.1090 (0.166)	
Firm Size	−0.1404** (0.014)	−0.0799 (0.104)	−0.0602 (0.135)	
Tangibility	−0.1117* (0.058)	−0.1795*** (0.000)	−0.1014*** (0.007)	
Sales Growth	0.0532** (0.047)	0.0967*** (0.000)	0.0950*** (0.000)	
Liquidity	0.0044 (0.818)	−0.0053 (0.473)	0.0055 (0.428)	

The main findings hold across different subsamples. Pre-COVID data shows some non-linear effects, while post-COVID results confirm a **strong negative linear relationship**. This suggests that changing macro conditions, especially higher interest rates, have made debt more burdensome.

7. FINDINGS AND DISCUSSION

7.1 Findings

Finding 1 — Leverage has a significant and negative effect on firm performance

The Two-Way Fixed Effects results clearly show that financial leverage has a statistically significant negative impact on the Return on Assets of Indian manufacturing firms ($\beta = -0.1679$, $p = 0.001$). In simple terms, a one-unit increase in the debt-to-asset ratio leads to a

16.79 percentage point drop in ROA, even after accounting for firm-specific effects, time effects, firm size, tangibility, sales growth and liquidity.

The relationship between leverage and firm performance is not neutral — it is clearly negative and economically meaningful across the full sample of 78 firms from 2016 to 2025.

Finding 2 — The non-linear hypothesis is not supported in the full sample

The squared leverage term (LEV^2) has a negative coefficient of -0.0778 but is not statistically significant ($p = 0.307$) in the main TWFE model. It also remains insignificant in two out of the three robustness checks.

This means that the expected inverted-U relationship suggested by Trade-off Theory — where firms benefit from debt up to an optimal point — is not observed in this dataset. Instead, the relationship appears to be consistently negative without any turning point.

There is no clear evidence of an optimal leverage level within the observed range, and the leverage-performance relationship does not follow a hump-shaped pattern.

Finding 3 — Sales growth is the most consistent and dominant driver of firm profitability

Among all the control variables, sales growth stands out as the strongest and most consistent driver of ROA. It is positive and highly significant across every model, including the main TWFE results ($\beta = 0.0810$, $p < 0.001$), the pre-COVID period ($\beta = 0.0532$, $p = 0.047$), the post-COVID period ($\beta = 0.0967$, $p < 0.001$), and the outlier-restricted sample ($\beta = 0.0950$, $p < 0.001$).

Tangibility is consistently negative and significant, while liquidity remains insignificant once leverage is controlled for.

It suggests that **revenue growth and asset structure matter more for profitability than capital structure decisions**, at least within this sample.

Finding 4 — A structural shift in the leverage-performance relationship occurred post-COVID

The sub-period analysis reveals a clear change in the relationship around the COVID-19 period.

The Non-Linear Impact Of Leverage On Firm Performance: A Panel Data Analysis Of Indian Manufacturing Firms

In the pre-COVID years (2016–2019), leverage itself is not significant ($\beta = 0.0279$, $p = 0.763$), but its squared term is negative and significant ($\beta = -0.2692$, $p = 0.047$). This indicates a partial non-linear or concave relationship under relatively stable economic conditions.

However, in the post-COVID period (2020–2025), the pattern changes sharply. Leverage becomes significantly negative ($\beta = -0.2084$, $p = 0.033$), while the squared term becomes completely insignificant ($p = 0.971$). In other words, the relationship shifts to a clearly negative linear form.

The leverage-performance relationship did change structurally after COVID, likely due to higher borrowing costs and tighter financial conditions.

Finding 5 — The negative leverage effect is robust and not driven by outliers

When the analysis is restricted to firms in the middle 90% of the size distribution, the main finding remains unchanged. The leverage coefficient stays negative and significant ($\beta = -0.1712$, $p = 0.002$), while LEV^2 is again insignificant ($p = 0.166$).

Since the results hold across the full sample, the post-COVID sub-period and the outlier-restricted sample, it can be concluded that the negative impact of leverage is not driven by extreme observations or a specific time frame.

The findings are stable and carry broader policy relevance across firms of different sizes.

7.2 Discussion

Discussion Point 1 — Stronger support for Pecking Order Theory than Trade-off Theory

The main result — a consistently negative and significant relationship between leverage and performance, with no evidence of a meaningful quadratic effect — lines up more closely with the Pecking Order Theory proposed by Myers (1984) and Myers and Majluf (1984).

According to this view, firms with higher leverage are often those that have already exhausted internal funds, which may signal weaker financial health rather than efficient use of debt. This interpretation fits well with the data, where the average leverage is only 15.56%, yet even at these levels, the negative impact on performance is already visible.

The Non-Linear Impact Of Leverage On Firm Performance: A Panel Data Analysis Of Indian Manufacturing Firms

The absence of an optimal leverage point also contrasts with the predictions of Modigliani and Miller (1963) and the empirical findings of Margaritis and Psillaki (2010). It suggests that in the Indian context, the benefits of the tax shield may not be strong enough to offset the costs of debt in the same way they might in more developed financial markets.

Discussion Point 2 — Consistency with recent India-specific studies

The results are broadly in line with more recent studies in the Indian manufacturing context. Panchal and Chand (2023) also find a negative relationship between leverage and ROA, and Pal (2022) shows that this negative effect becomes stronger during periods of tighter credit.

This study mirrors those findings, especially in the post-COVID period where the magnitude of the negative coefficient increases noticeably.

At the same time, this study goes a step further by testing for non-linearity and identifying a clear structural break after COVID, which earlier studies did not explore. This makes the contribution more methodologically and contextually relevant.

Discussion Point 3 — Explaining differences with earlier Indian evidence

The findings differ to some extent from earlier studies like Chadha and Sharma (2015), who found no significant effect, and Singh and Bagga (2019), who reported a positive relationship.

There are a few reasons for these differences. First, the time periods studied are different. Earlier periods were characterised by lower interest rates and easier access to credit, which may have reduced the cost of borrowing. Second, Singh and Bagga (2019) included firms from multiple sectors, which may dilute effects that are more specific to manufacturing. Third, earlier studies often used simpler estimation techniques, whereas this study uses a more rigorous Two-Way Fixed Effects approach with clustered errors.

This highlights an important point made by Harris and Raviv (1991): results in capital structure research are highly sensitive to the data, time period and methodology used.

Discussion Point 4 — The role of macroeconomic conditions and interest rates

The structural shift observed after COVID adds an important dimension to the discussion. It shows that the effect of leverage is not constant, but depends on broader economic conditions.

The Non-Linear Impact Of Leverage On Firm Performance: A Panel Data Analysis Of Indian Manufacturing Firms

This finding builds on the idea of conditional effects suggested by Ibhagui and Olokoyo (2018), extending it from firm characteristics to macroeconomic conditions. The increase in RBI repo rates between May 2022 and February 2023 raised borrowing costs, making debt more expensive for firms.

As a result, the balance between the benefits and costs of leverage shifts, weakening the logic of Trade-off Theory. This aligns with Jensen and Meckling (1976), who argue that the benefits of debt depend on the broader cost environment.

Discussion Point 5 — Understanding the negative role of tangibility in context

The consistently negative coefficient on tangibility deserves closer attention. Traditional theory, such as Rajan and Zingales (1995), views tangibility as beneficial because it provides collateral and improves access to debt.

However, the fixed effects approach used here captures changes within firms over time. From this perspective, increasing asset levels may actually reduce short-term profitability due to higher depreciation costs, maintenance expenses and delays in generating returns from new investments.

This is particularly relevant in the 2020–2025 period, when many Indian firms expanded capacity under the Production-Linked Incentive (PLI) scheme. Such investments increase assets immediately but may take time to translate into higher revenues, temporarily lowering ROA.

This suggests that in the Indian manufacturing context, tangibility is better understood in terms of **investment timing and utilisation**, rather than just collateral value.

8. CONCLUSION

This study analysed how financial leverage affects firm performance for 78 BSE-listed Indian manufacturing firms over the period 2016 to 2025. Using a Two-Way Fixed Effects panel model with firm-level clustered standard errors, the results show that leverage has a clear and statistically significant negative impact on Return on Assets. At the same time, the squared leverage term remains insignificant in the full sample, which means there is no evidence of the

The Non-Linear Impact Of Leverage On Firm Performance: A Panel Data Analysis Of Indian Manufacturing Firms

inverted-U relationship predicted by Trade-off Theory. Instead, the findings point more strongly toward Pecking Order Theory, where higher leverage reflects financial pressure rather than value-enhancing financing decisions.

The sub-period analysis adds another important layer to these results. In the pre-COVID years, when credit conditions were relatively stable, there was some evidence of a non-linear relationship. However, this pattern disappears in the post-COVID period, where the relationship becomes clearly and consistently negative. This shift suggests that the impact of leverage in the Indian manufacturing sector is highly sensitive to changes in the interest rate environment.

Across all specifications, sales growth stands out as the most consistent and important driver of profitability. In contrast, increasing tangibility is associated with lower short-term returns, likely reflecting the effect of higher capital expenditure, especially during the expansion phase linked to the PLI scheme.

Overall, this study contributes to the existing literature in two ways. First, it provides direct evidence against the non-linear leverage hypothesis in the Indian manufacturing context using recent panel data. Second, it highlights a structural break in the leverage-performance relationship following COVID-19, showing how macroeconomic conditions can reshape firm-level financial dynamics.

These findings have practical implications. For financial managers, the results suggest that relying heavily on debt may not be beneficial, especially in periods of rising interest rates. Instead, focusing on improving sales performance and managing capital investments carefully appears more effective for sustaining profitability. For policymakers, the results underline how sensitive firms are to changes in borrowing costs. This points to the need for supportive credit policies, such as interest subsidies or long-term financing options, to help firms invest without hurting short-term performance.

There is also scope for future research. One possible extension would be to use dynamic panel methods, such as the Arellano–Bond GMM approach, to address potential endogeneity issues between leverage and performance. Another direction would be to formally test the structural break identified around the COVID-19 period using econometric techniques like Chow tests or threshold models. Finally, expanding the analysis across other sectors such as IT,

pharmaceuticals, and FMCG, and including governance-related variables like promoter ownership or board structure, could provide a more complete understanding of how capital structure decisions affect firm performance in the Indian context.

9. REFERENCES

- Chadha, S., & Sharma, A. K. (2015). Capital structure and firm performance: Empirical evidence from India. *Vision: The Journal of Business Perspective*, 19(4), 295–302.
- Harris, M., & Raviv, A. (1991). The theory of capital structure. *The Journal of Finance*, 46(1), 297–355.
- Ibhagui, O. W., & Olokoyo, F. O. (2018). Leverage and firm performance: New evidence on the role of firm size. *North American Journal of Economics and Finance*, 45, 57–82.
- Jensen, M. C. (1986). Agency costs of free cash flow, corporate finance, and takeovers. *The American Economic Review*, 76(2), 323–329.
- Jensen, M. C., & Meckling, W. H. (1976). Theory of the firm: Managerial behavior, agency costs and ownership structure. *Journal of Financial Economics*, 3(4), 305–360.
- Margaritis, D., & Psillaki, M. (2010). Capital structure, equity ownership and firm performance. *Journal of Banking & Finance*, 34(3), 621–632.
- Modigliani, F., & Miller, M. H. (1958). The cost of capital, corporation finance and the theory of investment. *The American Economic Review*, 48(3), 261–297.
- Modigliani, F., & Miller, M. H. (1963). Corporate income taxes and the cost of capital: A correction. *The American Economic Review*, 53(3), 433–443.
- Myers, S. C. (1984). The capital structure puzzle. *The Journal of Finance*, 39(3), 575–592.
- Myers, S. C., & Majluf, N. S. (1984). Corporate financing and investment decisions when firms have information that investors do not have. *Journal of Financial Economics*, 13(2), 187–221.

The Non-Linear Impact Of Leverage On Firm Performance: A Panel Data Analysis Of Indian Manufacturing Firms

Pal, S. (2022). Influence of capital structure on firm performance: Empirical evidence from Indian manufacturing industry. *International Journal of Business Management and Finance Research*, 5(2), 100–110.

Panchal, S., & Chand, S. (2023). Impact of capital structure on firm performance: Empirical evidence from Indian manufacturing sector. *Journal of Asia Business Studies*, 17(5), 1063–1080.

Rajan, R. G., & Zingales, L. (1995). What do we know about capital structure? Some evidence from international data. *The Journal of Finance*, 50(5), 1421–1460.

Singh, N. P., & Bagga, M. (2019). The effect of capital structure on profitability: An empirical panel data study. *Jindal Journal of Business Research*, 8(1), 65–77.

Zeitun, R., & Tian, G. G. (2007). Capital structure and corporate performance: Evidence from Jordan. *Australasian Accounting, Business and Finance Journal*, 1(4), 40–61.